

P6-M: Schedule Topology under a Frozen English-Tagalog Token-Share Bundle (#307969)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

Holding the English-Tagalog source-content token quotas, tokenizer, parent type, training budget, and optimizer-reset policy fixed, does changing only the phase-two schedule topology-fine alternating, coarse alternating, blocked Tagalog-then-English, or one pinned randomized schedule-change the preregistered Tagalog and English validation BPB contrasts? This is a fixed one-seed mechanism comparison, not a population estimate.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent variables are full-split validation bits per byte (val_bpb_full ; lower is better) on the frozen Tagalog and English validation sets, measured using the pinned evaluator, carry-forward tokenizer, context length 2048, and fixed one-pass packing policy.

For each $\tau \in \{M\text{-coarse}, M\text{-blocked}, M\text{-rand}\}$:

$$\Delta TL(\tau) = TL_BPB(M\text{-}\tau) - TL_BPB(M\text{-fine})$$

$$\Delta EN(\tau) = EN_BPB(M\text{-}\tau) - EN_BPB(M\text{-fine})$$

Secondary contextual contrasts are:

$$R_TL(\tau) = TL_BPB(M\text{-}\tau) - TL_BPB(C2)$$

$$A_EN(\tau) = EN_BPB(M\text{-}\tau) - EN_BPB(C1)$$

All six child conditions $\times$ two languages remain lockedbox until one formal unblinding. The M-fine restricted-test values are descriptive secondary outcomes excluded from topology comparisons.

4) How many and which conditions will participants be assigned to?

No human participants. One seed-4 C0 parent will generate six sibling child conditions: C1, C2, M-fine, M-coarse, M-blocked, and M-rand.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

After the six named child conditions $\times$ two validation languages are sealed, one frozen script will perform a single formal unblinding.

For each $\tau \in \{M\text{-coarse}, M\text{-blocked}, M\text{-rand}\}$, the six primary contrasts are $\Delta TL(\tau)$ and $\Delta EN(\tau)$ as defined in Question 3. Using $\delta = 0.01$ BPB, each will be classified as lower (≤ -0.01), higher ($\geq +0.01$), or no material difference ($-0.01 < \Delta < +0.01$).

For all four mixed topologies, $R_TL(\tau)$ and $A_EN(\tau)$ will be reported as descriptive secondary contextual contrasts. Every finite terminal value will be retained. No p-values, confidence intervals, population estimates, across-seed averages, adaptive topology selection, or P5-style recurrence counts will be calculated. The M-fine restricted-test values will be reported separately and cannot alter any topology classification.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

No finite terminal BPB value will be treated as an outlier or excluded because of its magnitude or direction. No winsorization, seed replacement, or arm replacement will occur.

If the seed-4 parent fails the preregistered P0-T eligibility gate, no children will be trained and the study will close as blocked. Technical recovery is governed by P6-M-PREFILING-ADDENDUM.md, SHA-256 df49664809ada69d23dcd2c799e75b30f0fdb9afd7aee12b071ac24ef2f81082, and is permitted only for documented infrastructure failures before protected outcome access. NaN/Inf, unauthorized access, mismatched inputs, or unrecoverable terminal artifacts will be reported under the filed failure taxonomy rather than replaced or selectively rerun.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

One fixed parent-initialization seed (4) will produce one eligible C0 parent and six sibling phase-two conditions. The primary dataset will contain exactly 12 aggregate validation observations: two languages for each of the six child conditions.

If P0-T passes, all six children will be completed; no additional seeds or replacement arms will be added. Policy A permits one M-fine restricted-test event, producing two descriptive secondary language-level values. Each child receives exactly 294 updates (19,267,584 model-visible tokens). In-loop monitoring is diagnostic only and cannot determine sample size, arm continuation, checkpoint selection, or model selection.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual

analyses planned?)

This prospective P6-M study is informed by released P1.1-P5 evidence, including P5 #307836, but no P6-M seed-4 parent, child, validation scalar, restricted-test scalar, topology comparison, or P6-specific model-selection outcome has been collected or viewed before filing. P6-M is prospective but not outcome-independent.

After filing, authority is the filed PDF; the prefilling addendum at SHA-256

df49664809ada69d23dcd2c799e75b30f0fdb9afd7aee12b071ac24ef2f81082; the unmodified gate plan at SHA-256

d8a63608608c59d2c4d9882e5346625462056c331094942a8a01d496697a1c79; then the P6 LOCK, manifests, and deterministic receipts. The addendum binds topology manifest SHA-256 d1e7d5af7247a572e319ee003b5f4e3da5d1fb1592e5ed9ff6b22eeec15ea606.

All mixed arms use exact source-content quotas of 9,633,792 Tagalog and 9,633,792 English tokens, identical within-language streams, stored no-wrap schedules, and no runtime reshuffling. M-fine uses 2,048-token EN-first alternating blocks; M-coarse uses 1,204,224-token EN-first alternating blocks; M-blocked is all Tagalog then all English; M-rand is one precomputed sequence containing 4,704 blocks per language, shuffled once using Python random.Random(42) and hash-pinned before filing.

All six child conditions × two languages are sealed before exactly one authorized M-fine-only secondary restricted-test event and one formal unblinding. C1, C2, M-coarse, M-blocked, and M-rand are not tested, and no cross-topology test comparison is computed.

Technical recovery follows the addendum's pre-outcome taxonomy from immutable inputs. Valid terminal checkpoints are accepted without retraining; outcome-bearing or unrecoverable protocol stops are reported rather than replaced. P6-M measures BPB schedule topology, not downstream Filipino-task performance or all Philippine languages.

BUNDLE

This pre-registration is part of a bundle which includes:

#307,591 - <https://aspredicted.org/if84km.pdf> - Title: 'P4: token-share mix after fresh TL parent (nanochat, TL-39/WikiText-103)' #306,780 -

<https://aspredicted.org/6r6v4v.pdf> - Title: 'NANOCHAT-FILIPINO P1.1 - WikiText-TL-39 fixed-budget depth vs held-out BPB' #307,342 -

<https://aspredicted.org/wd2pc8.pdf> - Title: 'P3: TL retention after EN continuation (nanochat, TL-39 then WikiText-103)' #306,935 -

<https://aspredicted.org/xa56bs.pdf> - Title: 'P2: EN retention after TL continuation (nanochat, WikiText-103 then TL-39)' #307,836 -

<https://aspredicted.org/k6ib64.pdf> - Title: 'P5: multi-seed panel of the P4 token-share mix after fresh TL parents'